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Inventor(s)	Best; Michael J. et al.

Radome with ceramic matrix composite

Abstract

In a first example, a radome includes a shell including a ceramic matrix composite, the shell forming a first hole at a forward end of the shell and a second hole at an aft end of the shell. The radome also includes a fluid impervious coating on the shell. In a second example, a vehicle includes a main body, the radome, and an attachment assembly that couples the radome to the main body. In a third example, a method includes forming a shell comprising a ceramic matrix composite using a wet layup process, applying a fluid impervious coating onto the shell, and curing the shell and the fluid impervious coating.

Inventors: **Best; Michael J. (Chicago, IL), DiChiara; Robert A. (Chicago, IL), Muench; Brian L. (Chicago, IL), Heng; Vann (Chicago, IL), Embler; Jonathan D. (Chicago, IL), Pinney; Thomas R. (Chicago, IL), O'Brien; Holly M. (Chicago, IL)**

Applicant: **The Boeing Company** (Arlington, VA)

Family ID: **86769113**

Assignee: **The Boeing Company (Arlington, VA)**

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Primary Examiner: Levi; Dameon E

Assistant Examiner: DeWitt; Jordan E.

Attorney, Agent or Firm: Hanley, Flight & Zimmerman, LLC

Background/Summary

CROSS REFERENCE TO RELATED APPLICATION (1) This application is a divisional of U.S. application Ser. No. 17/553,238, filed on Dec. 16, 2021, the entire contents of which are hereby incorporated by reference herein.

FIELD
(1) The present disclosure generally relates to a radome, and more specifically to a radome formed with a ceramic matrix composite.
BACKGROUND
(2) A radome is a structural enclosure that protects a radar antenna. For high speed vehicles (e.g., aircraft) that have a radar antenna, radomes are often formed of silicon nitride (Si.sub.3N.sub.4), which has several limitations. First, current processes for shaping silicon nitride are generally monolithic (e.g., void of reinforcement techniques), making the material more brittle for handling, machining, and drilling holes for attachments. Thus, machining Si.sub.3N.sub.4 materials is costly and has a practical limit for how large of a component can be manufactured, which limits design flexibility of radomes. Second, Si.sub.3N.sub.4 radomes are typically bonded to a vehicle with glass or ceramic adhesive which is a complex process. Such bond joints are typically permanent in nature, which hampers the ability to repair a radar antenna within the radome or to inspect the interior of the radome. Also, Si.sub.3N.sub.4 is brittle, vulnerable to thermal shock incurred during high speed travel, and has higher transmission loss for radio frequencies (RF) utilized by the radar antenna when the vehicle travels at high speeds that heat the Si.sub.3N.sub.4.
SUMMARY
(3) One aspect of the disclosure is a radome comprising a shell comprising a ceramic matrix composite, the shell forming a first hole at a forward end of the shell and a second hole at an aft end of the shell; and a fluid impervious coating on the shell.
(4) Another aspect of the disclosure is a vehicle comprising: a main body; a radome comprising: a shell comprising a ceramic matrix composite, the shell forming a first hole at a forward end of the shell and a second hole at an aft end of the shell; and a fluid impervious coating on the shell; and an attachment assembly that couples the radome to the main body.
(5) Another aspect of the disclosure is a method of forming a radome, the method comprising: forming a shell comprising a ceramic matrix composite using a wet layup process; applying a fluid impervious coating onto the shell; and curing the shell and the fluid impervious coating.
(6) By the term “about” or “substantially” with reference to amounts or measurement values described herein, it is meant that the recited characteristic, parameter, or value need not be achieved exactly, but that deviations or variations, including for example, tolerances, measurement error, measurement accuracy limitations and other factors known to those of skill in the art, may occur in

amounts that do not preclude the effect the characteristic was intended to provide.
(7) The features, functions, and advantages that have been discussed can be achieved independently in various examples or may be combined in yet other examples further details of which can be seen with reference to the following description and drawings.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
(1) The novel features believed characteristic of the illustrative examples are set forth in the appended claims. The illustrative examples, however, as well as a preferred mode of use, further objectives and descriptions thereof, will best be understood by reference to the following detailed description of an illustrative example of the present disclosure when read in conjunction with the accompanying Figures.
(2) FIG. **1** is a perspective view of a forward end of a radome, according to an example.
(3) FIG. **2** is a perspective view of an aft end of a radome, according to an example.
(4) FIG. **3** is a side view of a radome, according to an example.
(5) FIG. **4** is a perspective view of a forward end of a radome, according to an example.
(6) FIG. **5** is a perspective view of a forward end of a radome, according to an example.
(7) FIG. **6** is a cross sectional view of a radome and an attachment assembly, according to an example.
(8) FIG. **7** is an exploded view of a fastening assembly and a tip, according to an example.
(9) FIG. **8** is a close up cross sectional view of a radome, according to an example.
(10) FIG. **9** is a perspective view of an aft end of a radome, according to an example.
(11) FIG. **10** is a perspective view of attachment assemblies, according to an example.
(12) FIG. **11** is an exploded view of a radome and an attachment assembly, according to an example.
(13) FIG. **12** is a cross sectional view of a radome and an attachment assembly, according to an example.
(14) FIG. **13** is a perspective view of a radome and a vehicle, according to an example.
(15) FIG. **14** is a cross sectional view of a radome and a vehicle, according to an example.
(16) FIG. **15** is a cross sectional view of a radome and a vehicle, according to an example.
(17) FIG. **16** is a block diagram of a method, according to an example.
DETAILED DESCRIPTION
(18) A need exists for a radome that is formed of a material that is easier and less costly to process, that can be more easily attached and removed from a vehicle, that has desirable transmission characteristics with respect to radio frequencies (RF), and is able to withstand harsh flight environments such as high heat and structural loads, rain or hail impact, tool drop, and handling. Examples herein include a radome that includes a shell including (e.g., formed of) a ceramic matrix composite (CMC). The shell forms a first hole at a forward end of the shell and a second hole at an aft end of the shell. The aft end of the shell can be placed over a radar antenna and attached to a vehicle. The radome also includes a fluid impervious coating on the shell to keep the radar antenna isolated from moisture, vapors, liquids (e.g., rain), or hot gases present during operation (e.g., hot gases generated via friction of the aircraft body with air during high speed flight).
(19) In some examples, the shell has an aerodynamic conical shape. When compared to Si.sub.3N.sub.4, the CMC is easier to form and process, and generally provides enhanced structural stability and exhibits stable RF transmission with low losses caused by temperature variations. Additionally, the CMC can be attached to a vehicle via fasteners to facilitate easier removal for maintenance and inspection.
(20) Disclosed examples will now be described more fully hereinafter with reference to the

accompanying Drawings, in which some, but not all of the disclosed examples are shown. Indeed, several different examples may be described and should not be construed as limited to the examples set forth herein. Rather, these examples are described so that this disclosure will be thorough and complete and will fully convey the scope of the disclosure to those skilled in the art.

(21) FIG. **1** is a perspective view of a forward end **106** of a radome **100** (e.g., of a shell **102**). The radome **100** includes the shell **102** that includes (e.g., is formed of) a ceramic matrix composite (CMC). The shell **102** forms a first hole **104** at the forward end **106** of the shell **102** and a second hole **108** at an aft end **110** of the shell **102**. The radome **100** also includes a fluid impervious coating on an outer surface **116** of the shell **102** (e.g., on an entirety of the outer surface **116**). The outer surface **116** can be defined as facing away from an axis **156** (e.g., a longitudinal axis of the shell **102** and of the radome **100**). The first hole **104** and the second hole **108** are aligned on the axis **156**. The shell **102** also includes holes **141** that are formed via drilling after formation of the shell **102**. The holes **141** are configured to receive respective fasteners as discussed below. The fluid impervious coating is shown in more detail in FIG. **6**, FIG. **8**, and FIG. **12**.

(22) As shown, the shell **102** has a hollow and somewhat conical shape. The first hole **104** is smaller than the second hole **108** (e.g., to form an aerodynamic shape of the shell **102** that can be placed over a radar antenna via the second hole **108**). The CMC includes ceramic fibers (e.g., ceramic oxide fibers) reinforced in a ceramic matrix (e.g., a ceramic oxide matrix). The ceramic fibers and the ceramic matrix can each include any ceramic material that is substantially RF transparent. For example, the ceramic fibers and the ceramic matrix can each include any materials that are (e.g., inorganic) non-metallic oxides such as alumina, silica, an alumina-silicate compound mixture, or cordierite. When compared to silicon nitride, the (e.g., oxide) CMC generally has a lower dielectric coefficient and is more transmissive for RF frequencies. The oxide CMC radome materials have a substantially stable dielectric constant of approximately 5 and a stable low loss tangent (e.g., <0.01 or <0.02) from 2 GHz to 40 GHz. The dielectric constant of the oxide CMC materials is generally stable with respect to temperature, even up to temperatures of 1800° F. That is, a signal transmitted or received by a radar antenna under the radome **100** will typically not be significantly attenuated as the signal passes through the radome **100**. The materials of the shell **102** generally have a dielectric constant of 5 or 6 or less.

(23) One way of forming the CMC is to infiltrate a ceramic precursor solution that includes fine ceramic particles into oxide ceramic fibers preform to produce ceramic prepreg. The ceramic prepreg can be formed on a tool surface or mold having the geometry of the shell **102**. Additional details regarding how the radome **100** is formed, including details about the fluid impervious coating, are discussed below with reference to FIG. **6**.

(24) FIG. **2** is a perspective view of the aft end **110** of the radome **100** (e.g., of the shell **102**). As shown, the aft end **110** forms the largest diameter of the radome **100** (e.g., of the shell **102**) with respect to the axis **156**. The aft end **110** has an annular shape. Also shown in FIG. **2** is an inner surface **114** of the shell **102**. The inner surface **114** can be defined as facing toward the axis **156**.

(25) FIG. **3** is a side view of the radome **100**.

(26) FIG. **4** is a perspective view of the forward end **106** of the radome **100**. A tip **118** of the radome **100** is shown disassembled from the shell **102**. The tip **118** is formed of a ceramic material, such as silicon nitride or alumina. Using such materials for the tip **118** can provide similar thermal expansion properties for the tip **118** and the shell **102**.

(27) FIG. **5** is a perspective view of the forward end **106** of the radome **100**, showing the tip **118** placed within the first hole **104** to form a fluid tight seal **120** with the shell **102** over the first hole **104** (e.g., to prevent fluid from leaking into the shell **102** via the first hole **104**). The tip **118** also forms an aerodynamic surface with the shell **102**.

(28) FIG. **6** is a cross sectional view of the radome **100**. In particular, FIG. **6** shows the inner surface **114** of the shell **102**, the outer surface **116** of the shell **102**, and the fluid impervious coating **112**. The fluid impervious coating **112** can be a sintered glass layer with transmission properties

similar to that of the material of the shell **102**. As shown, the fluid impervious coating **112** covers an entirety of the outer surface **116**. The radome **100** also includes an attachment assembly **132** configured to couple the shell **102** to a vehicle **200** (discussed in more detail below). In FIG. **6**, the thickness of the fluid impervious coating **112** is exaggerated for purposes of illustration. In practice, the thickness of the fluid impervious coating **112** is generally much smaller than a thickness of the shell **102** (e.g., a distance between the outer surface **116** and the inner surface **114**). For example, the thickness of the shell **102** generally ranges from 0.1 inches to 0.25 inches and the thickness of the fluid impervious coating **112** generally ranges from 0.005 to 0.01 inches.
(29) The shell **102** can be formed by using a wet layup process. That is, a stack of ceramic prepreg including the ceramic fibers and ceramic matrix can be placed over a conical mold to form a desired radial thickness of the CMC over the mold (e.g., a forming tool). The stacked ceramic prepregs are consolidated to each other via vacuum and autoclave pressure during cure while on the conical mold and heated to an intermediate temperature (e.g., 180-350° F.) until a solid structure is formed (e.g., the shell **102**). Then, the shell **102** is removed from the mold and processed and cured to a high temperature (e.g., 350-2000° F.) in a furnace. The shell **102** can then be machined to fine tune the dimensions of the shell **102** as desired.
(30) After the shell **102** is formed, there are at least two ways to apply the fluid impervious coating **112** onto the outer surface **116** of the shell **102**. One way is to apply the fluid impervious coating **112** after the autoclave cure of the shell **102** (e.g., at intermediate temperature) and then co-process the fluid impervious coating **112** and the shell **102** to a high temperature in a furnace. Another way is to apply the fluid impervious coating **112** after the high-temperature post processing of the shell **102** and repeat the high-temperature post processing for the fluid impervious coating **112**. For example, the fluid impervious coating **112** can be brushed onto or sprayed onto the shell **102**. The fluid impervious coating **112** can also be machined to reduce a thickness **208** of the fluid impervious coating **112** after the curing.
(31) FIG. **7** is an exploded view of a fastening assembly **122** and the tip **118**. The radome **100** includes the fastening assembly **122** that mechanically fastens the tip **118** to the shell **102** as is shown in FIG. **8**. The fastening assembly **122** is configured to receive an aft end **126** of the tip **118**. The aft end **126** generally includes male threads. The fastening assembly **122** includes a bushing **124**, a spring element **130** (e.g., a Belleville washer), a washer **131**, and a fastener **128** (e.g., a nut having female threads that mate with the male threads of the aft end **126**).
(32) FIG. **8** is a close up cross sectional view of the radome **100**. The fastening assembly **122** includes the bushing **124** that conforms to the inner surface **114** of the shell **102** (e.g., to prevent radial movement of the aft end **126** of the tip **118**). That is, the bushing **124** includes a radially outward facing surface that has the same shape as the inner surface **114**. The bushing **124** is configured to receive the aft end **126** of the tip **118**. Also, the fastener **128** is configured to mate with the aft end **126** of the tip **118** to hold the tip **118** against the shell **102** over the first hole **104**. The spring element **130** is positioned between the fastener **128** and the bushing **124** (e.g., between the washer **131** and the bushing **124**) and is configured to receive the aft end **126** of the tip **118**. The spring element **130** is also configured to flex to maintain a preload on the shell **102** and/or the tip **118** during thermal expansion of the shell **102** or during thermal contraction of the shell **102**.
(33) Starting at the tip **118**, the thickness **208** of the fluid impervious coating **112** is shown to increase moving in the aft direction. However, this is a result of the exaggerated thickness **208** shown in FIG. **8** and is not necessarily present in practice.
(34) FIG. **9** is a perspective view of the aft end **110** of the radome **100**, showing the attachment assembly **132**.
(35) FIG. **10** is a perspective view of attachment assemblies **132**. The attachment assembly **132** on the left is formed of a singular piece of material (e.g., a metal such as a titanium alloy, or a superalloy such as Inconel® 718) whereas the attachment assembly **132** on the right is formed of eight separate pieces (e.g., metal). The attachment assembly **132** on the left might be stronger than

the attachment assembly **132** on the right, however, the attachment assembly **132** on the right might be lighter than the attachment assembly **132** on the left.

(36) Both attachment assemblies **132** include an annular component **136** configured to be attached to the vehicle **200** (e.g., via fasteners) and a bipod component **138** configured to attach the annular component **136** to the shell **102**. The bipod components **138** are generally machined (e.g., integrally formed with the annular component **136** via subtractive manufacturing). The annular component **136** forms several holes **140** that are each configured to receive a fastener that attaches the annular component **136** to the vehicle **200**. Each bipod component **138** also forms a hole **143** configured to receive a fastener that attaches the bipod component **138** (e.g., the annular component **136**) to the shell **102**.

(37) In particular, the bipod component **138** includes a joint **146** that forms a hole **143** configured to receive a fastener and a first leg **148** that couples the joint **146** to a first attachment point **150** on the annular component **136**. The bipod component **138** also includes a second leg **152** that couples the joint **146** to a second attachment point **154** on the annular component **136**. The first leg **148** is machined to form the joint **146** with the second leg **152**. The first attachment point **150** and the second attachment point **154** are machined joints between the annular component **136** and the first leg **148** or the second leg **152**, respectively.

(38) FIG. **11** is an exploded view of the radome **100**, fasteners **142**, and the attachment assembly **132**. The fasteners **142** can be inserted through holes **141** in the shell **102** and holes **143** in the joints **146** to secure the attachment assembly to the shell **102**.

(39) FIG. **12** is a cross sectional view of the radome **100** and the attachment assembly **132**. The attachment assembly **132** is attached to the shell **102** via fasteners **142** (e.g., bolts or screws). For example, the fasteners **142** are inserted through (e.g., drilled) holes **141** in the shell **102** and through the holes **143** in the joint **146** of the bipod component **138**. In some examples, the fluid impervious coating **112** surrounds and covers holes **141** within the shell **102** and surrounds and covers several fasteners **142** that fasten the shell **102** to respective joints **146** through the holes **141** in the shell **102**. In some examples, the fluid impervious coating **112** extends into the holes **141** (e.g., through an entirety of each of the holes **141** in the shell **102**).

(40) The bipod component **138** generally exhibits more flexibility perpendicular to the axis **156** (e.g., in the radial direction) than parallel to the axis **156** (e.g., in the longitudinal direction) which will help accommodate potential thermal expansion of the shell **102**.

(41) The radome **100** also includes a fastening assembly **123** that includes a spring element **130** (e.g., a Belleville washer) that is configured to maintain a preload on the shell **102** and the bipod component **138** during thermal expansion of the shell **102** or during thermal contraction of the shell **102**.

(42) FIG. **13** is a perspective view of the radome **100** and the vehicle **200**. The vehicle **200** includes a main body **202**, the radome **100**, and the attachment assembly **132** that couples the radome **100** to the main body **202**. As shown, the vehicle **200** can be a commercial airliner with a nose mounted radar antenna covered by the radome **100**, but other examples are possible, such as an automobile, a boat, or an unmanned aerial vehicle (UAV).

(43) FIG. **14** is a cross sectional view of the radome **100** and the main body **202** of the vehicle **200**. As shown, fasteners **142** are inserted through the holes **140** in the annular component **136** to attach the attachment assembly **132** to the main body **202** of the vehicle **200**. The vehicle **200** also includes a metallic gasket **204** that forms a fluid impervious seal **206** between the radome **100** and the main body **202** (e.g., to prevent fluid leakage into the shell **102**). FIG. **14** shows an example where the metallic gasket **204** is tensioned in the radial direction.

(44) FIG. **15** is also a cross sectional view of the radome **100** and the main body **202** of the vehicle **200**. FIG. **15** shows an example where the metallic gasket **204** is tensioned in the longitudinal direction between a retainer **151** and the main body **202**, but nonetheless forms the fluid impervious seal **206** between the radome **100** (e.g., the retainer **151**) and the main body **202**.

(45) FIG. **16** is a block diagram of a method **300** for forming a radome. As shown in FIG. **16**, the method **300** includes one or more operations, functions, or actions as illustrated by blocks **302**, **304**, **306**, and **308**. Although the blocks are illustrated in a sequential order, these blocks may also be performed in parallel, and/or in a different order than those described herein. Also, the various blocks may be combined into fewer blocks, divided into additional blocks, and/or removed based upon the desired implementation.
(46) At block **302**, the method **300** includes forming the shell **102** comprising the ceramic matrix composite using a wet layup process. Block **302** is described above with respect to FIG. **6**.
(47) At block **304**, the method **300** includes applying the fluid impervious coating **112** onto the shell **102**. Block **304** is also described above with respect to FIG. **6**.
(48) At block **306**, the method **300** includes curing the shell **102** and the fluid impervious coating **112**. Block **306** is also described above with respect to FIG. **6**.
(49) At block **308**, the method **300** includes machining the fluid impervious coating **112** to reduce the thickness **208** of the fluid impervious coating **112** after the curing. Block **308** is also described above with respect to FIG. **6**.
(50) Examples of the present disclosure can thus relate to one of the enumerated clauses (ECs) listed below. EC 1 A radome comprising: a shell comprising a ceramic matrix composite, the shell forming a first hole at a forward end of the shell and a second hole at an aft end of the shell; and a fluid impervious coating on the shell. EC 2 The radome of EC 1, wherein the shell comprises an inner surface and an outer surface, and wherein the fluid impervious coating covers an entirety of the outer surface. EC 3 The radome of any of ECs 1-2, wherein the first hole is smaller than the second hole. EC 4 The radome of any of ECs 1-3, further comprising a tip that forms a fluid tight seal with the shell over the first hole. EC 5 The radome of EC 4, wherein the tip comprises a ceramic material. EC 6 The radome of any of ECs 4-5, further comprising a fastening assembly that mechanically fastens the tip to the shell. EC 7 The radome of EC 6, wherein the fastening assembly comprises: a bushing that conforms to an inner surface of the shell, wherein the bushing is configured to receive an aft end of the tip; and a fastener configured to mate with the aft end of the tip to hold the tip against the shell over the first hole. EC 8 The radome of EC 7, further comprising a spring element between the fastener and the bushing, wherein the spring element is configured to receive the aft end of the tip. EC 9 The radome of EC 8, wherein the spring element is configured to maintain a preload on the tip during thermal expansion of the shell or during thermal contraction of the shell. EC 10 The radome of any of ECs 1-9, further comprising an attachment assembly configured to couple the shell to a vehicle. EC 11 The radome of EC 10, wherein the attachment assembly comprises: an annular component configured to be attached to the vehicle; and a bipod component configured to attach the annular component to the shell. EC 12 The radome of EC 11, wherein the annular component forms a hole configured to receive a fastener. EC 13 The radome of any of ECs 11-12, wherein the bipod component forms a hole configured to receive a fastener. EC 14 The radome of any of ECs 11-13, wherein the bipod component comprises: a joint that forms a hole configured to receive a fastener; a first leg that couples the joint to a first attachment point on the annular component; and a second leg that couples the joint to a second attachment point on the annular component. EC 15 The radome of any of ECs 11-14, wherein the first hole and the second hole are aligned on an axis, and wherein the bipod component is more flexible perpendicular to the axis than parallel to the axis. EC 16 The radome of any of ECs 11-15, further comprising a fastening assembly comprising a spring element that is configured to maintain a preload on the shell and the bipod component during thermal expansion of the shell or during thermal contraction of the shell. EC 17 A vehicle comprising: a main body; a radome comprising: a shell comprising a ceramic matrix composite, the shell forming a first hole at a forward end of the shell and a second hole at an aft end of the shell; and a fluid impervious coating on the shell; and an attachment assembly that couples the radome to the main body. EC 18 The vehicle of EC 17, further comprising a metallic gasket that forms a fluid impervious seal between the radome and the main

body. EC 19 A method of forming a radome, the method comprising: forming a shell comprising a ceramic matrix composite using a wet layup process; applying a fluid impervious coating onto the shell; and curing the shell and the fluid impervious coating. EC 20 The method of EC 19, further comprising machining the fluid impervious coating to reduce a thickness of the fluid impervious coating after the curing.

(51) The description of the different advantageous arrangements has been presented for purposes of illustration and description, and is not intended to be exhaustive or limited to the examples in the form disclosed. Many modifications and variations will be apparent to those of ordinary skill in the art. Further, different advantageous examples may describe different advantages as compared to other advantageous examples. The example or examples selected are chosen and described in order to explain the principles of the examples, the practical application, and to enable others of ordinary skill in the art to understand the disclosure for various examples with various modifications as are suited to the particular use contemplated.

Claims

1. A method of forming a radome, the method comprising: forming a shell comprising a ceramic matrix composite using a wet layup process; applying a fluid impervious coating onto the shell; curing the shell and the fluid impervious coating; and coupling the shell to a vehicle by: attaching an annular component to the vehicle; and attaching a bipod component to the shell at a joint of the bipod component, the joint forward of the annular component, wherein a first leg of the bipod component is adjoined to the annular component aft from the joint, and a second leg of the bipod component is adjoined to the annular component aft from the joint.
2. The method of claim 1, further comprising machining the fluid impervious coating to reduce a thickness of the fluid impervious coating.
3. The method of claim 1, wherein forming the shell comprises forming the shell such that the shell comprises an inner surface and an outer surface, and wherein applying the fluid impervious coating comprises applying the fluid impervious coating such that the fluid impervious coating covers an entirety of the outer surface.
4. The method of claim 1, wherein forming the shell comprises forming the shell such that the shell forms a first hole at a forward end of the shell and a second hole at an aft end of the shell.
5. The method of claim 4, wherein the first hole is smaller than the second hole.
6. The method of claim 4, further comprising positioning a tip such that the tip forms a fluid tight seal with the shell over the first hole.
7. The method of claim 6, wherein the tip comprises a ceramic material.
8. The method of claim 6, further comprising using a fastening assembly to mechanically fasten the tip to the shell.
9. The method of claim 8, wherein using the fastening assembly comprises: positioning a bushing such that the bushing conforms to an inner surface of the shell and receives an aft end of the tip; and mating a fastener with the aft end of the tip to hold the tip against the shell over the first hole.
10. The method of claim 9, further comprising positioning a spring element between the fastener and the bushing such that the spring element receives the aft end of the tip.
11. The method of claim 10, wherein the spring element maintains a preload on the tip during thermal expansion of the shell.
12. The method of claim 10, wherein the spring element maintains a preload on the tip during thermal contraction of the shell.
13. The method of claim 1, wherein the annular component forms a hole configured to receive a fastener.
14. The method of claim 1, wherein the bipod component forms a hole configured to receive a fastener.

15. The method of claim 1, wherein the bipod component comprises: a joint that forms a hole configured to receive a fastener, wherein the first leg couples the joint to a first attachment point on the annular component; and the second leg couples the joint to a second attachment point on the annular component.
16. The method of claim 4, wherein the first hole and the second hole are aligned on an axis, and wherein the bipod component is more flexible along a first direction that is perpendicular to the axis than a second direction that is parallel to the axis.
17. The method of claim 1, further comprising using a fastening assembly comprising a spring element that is configured to maintain a preload on the shell during thermal expansion of the shell or during thermal contraction of the shell.
18. The method of claim 1, further comprising using a fastening assembly comprising a spring element that is configured to maintain a preload on the bipod component during thermal expansion of the shell or during thermal contraction of the shell.
19. The method of claim 1, wherein the first and second legs converge at the joint.
20. The method of claim 1, wherein the first and second legs are machined and are attached to the annular component at first and second at first and second machine joints, respectively.